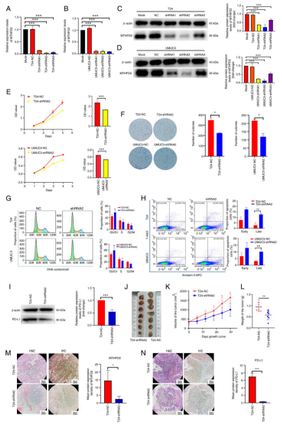
Panel-Level Fine-Grained Perception

Q: Observe Figure E. Which statement is incorrect?

A. In T24 cells, the proliferation curve of the shRNA2 group is consistently lower than that of the NC group.
B. In T24 cells, the gap between the proliferation curves of the shRNA2 group and the NC group increases over time.
C. The OD value of the UMUC3-shRNA2 group increases more slowly than that of the NC group.
D. The degrees of decrease in column height for the shRNA2 groups of both cell lines are similar.
E. Cannot be determined.

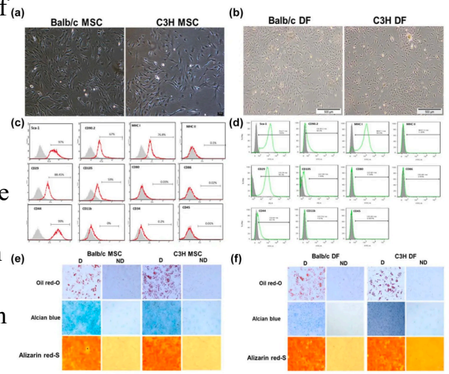


Answer: D

Numerical Perception (NP)

Q: Analyze the differentiation staining results of MSC in Figure e (Oil Red O, Alcian Blue, Alizarin Red S; D = differentiated, ND = non-differentiated). Which judgment is correct?

A. In MSC Oil Red O staining, the Balb/c ND group shows more lipid droplets than the C3H D group.
B. Comparing Alizarin Red S staining of Balb/c MSC and C3H MSC, the orange-red mineralization signals in the differentiated groups (D) are stronger than those in the non-differentiated groups (ND) in both.
C. Alizarin Red S staining of MSC shows that the Balb/c D group has more obvious orange mineralization than the C3H ND group.
D. After induced differentiation (D group), the positive signals of the three stainings (Oil Red O, Alcian Blue, Alizarin Red S) in Balb/c MSC are weaker than those in the corresponding ND groups of C3H MSC.
E. Cannot be determined.



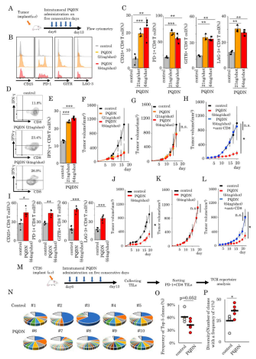
Answer: B

Morphological Perception (MP)

Cross-Panel Relation Understanding

Q: Comparing the trends of different doses of PQDN in Panel C, Panel E, and Panel F with the control group, which of the following visual perceptions is incorrect?

A. In Panel C, the molecular expression bars of the PQDN (64ng/shot) group are higher than those of the 21ng/shot group and exceed the control group, indicating a trend of higher expression with higher doses.
B. In Panel E, the $CD8^+$ T cell IFN-γ secretion bars in the PQDN treatment group are higher than in the control group, and the 64ng/shot group shows a greater increase than the 21ng/shot group, consistent with the trend in Panel C.
C. In Panel F, the tumor volume lines for the PQDN (64ng/shot) and 21ng/shot groups overlap due to error bars, visually suggesting no difference in tumor volume.
D. From Panel C to Panel E to Panel F, the PQDN (64ng/shot) group shows a coherent trend: higher molecular expression → higher cytokine secretion → slower tumor volume growth.
E. It is impossible to determine.

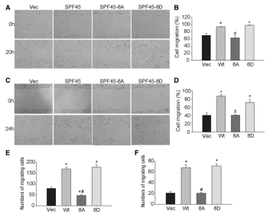


Answer: A

Trend Analysis (TA)

Q: Considering Panel A, Panel C, and their corresponding quantitative results in Panel B and D, which of the following descriptions best reflects the image data?

A. The scratch healing effect in the SPF45-8A group visually remains between the Vec group and the SPF45 group, and its quantitative migration rate also shows a similar intermediate level.
B. Whether at 20 h or 24 h, the scratch closure in the SPF45-8A group visually resembles that of the Vec group, and this observation is strongly supported by the similar quantitative migration percentages in Panel B and D.
C. Although the SPF45-8A group shows poorer scratch healing at 20 h, its cell migration approaches that of the SPF45 group at 24 h.
D. Panels A and C show that the cell morphology at the scratch edge in the SPF45-8A group appears more active and dispersed, suggesting enhanced migration ability, which visually contradicts the lower migration rates in Panel B and D.
E. It is impossible to determine.



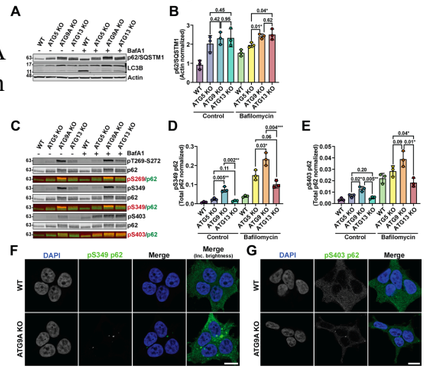
Answer: B

Heterogeneous Integration (HI)

Expert-Level Reasoning

Q: Based on the experimental images, which of the following is correct?

A. In Fig. A, ATG5 knockout leads to p62 accumulation, indicating blocked autophagic degradation independent of other ATG genes.
B. In Fig. B, p62 levels remain unchanged across ATG knockouts after BafA1 treatment, suggesting functional similarity.
C. Figs. C–E show phosphorylated p62 in ATG9A knockout, indicating its role in regulating p62 phosphorylation.
D. Figs. F and G show that ATG9A knockout does not affect p62 localization, suggesting no impact on subcellular distribution.
E. Cannot be determined.

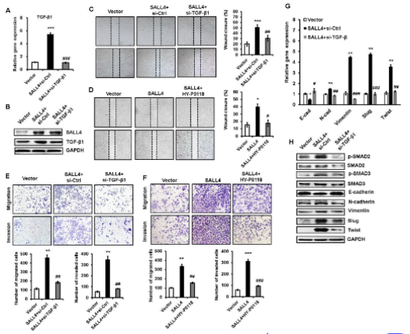


Answer: D

Qualitative Reasoning (Qual.)

Q: In SALL4-overexpressing HGC-27 cells (Fig. 5), which best describes key changes?

A. SALL4 raises TGF-β1 mRNA 6×, wound closure ≈ 60%, invasion 5×; si-TGF-β1 cuts migration >70%.
B. si-TGF-β1 lowers Slug, Vimentin, Twist mRNA 60%–80%, protein and p-SMAD2/3 decrease, showing TGF-β/SMAD inhibition.
C. HY-P0118 reduces wound closure 60%→20%, migration 450→100, p-SMAD2/3 returns to control, blocking TGF-β1.
D. si-TGF-β1 leaves N-cadherin, Vimentin mRNA unchanged, minor protein drop, E-cadherin stable; no EMT reversal.
E. Cannot be determined.



Answer: B

Quantitative Reasoning (Quant.)